

Kaiming Liu

Physics PhD student, University of Illinois Urbana-Champaign

kl54@illinois.edu · kl543.github.io · [GitHub](#) · [LinkedIn](#)

Research Interests

Scientific machine learning; inverse problems; probabilistic and multimodal inference; physics-based forward validation; simulation-to-experiment transfer; computational condensed matter physics.

Education

University of Illinois Urbana-Champaign

Aug 2023–present

Ph.D. in Physics, in progress

Advisor: Bryan K. Clark

Xi'an Jiaotong University

Sep 2019–Jul 2023

B.S. in Physics (Honors)

University of California, Berkeley

Aug–Dec 2021

Berkeley Physics International Education (BPIE) Program

Non-degree visiting program

Xi'an Jiaotong University

Sep 2017–Jul 2019

Special Class for the Gifted Young

Research Experience

University of Illinois Urbana-Champaign

Nov 2025–present

Graduate research, Bryan K. Clark group

- Develop probabilistic approaches to crystal electric field inverse problems, including multimodal parameter inference and reconstruction of observables with physics-based forward models.
- Study the sensitivity of scientific inference to experimental noise and simulation-to-experiment distribution shift, with applications to magnetic and scattering models.
- Implemented neural-network backflow wavefunctions for strongly correlated fermionic systems and ran variational Monte Carlo simulations with parallel sampling on computing clusters.
- Profiled sampling, local-energy evaluation, and optimization stages; refactored sampling routines to reuse wavefunction evaluations.

University of Illinois Urbana-Champaign

Aug 2023–Oct 2025

Research assistant, Vidya Madhavan group

- Grew thin-film heterostructures by molecular beam epitaxy and characterized materials using STM/STS in ultrahigh-vacuum systems.
- Built an automated MBE temperature-control program with feedback loops, measurement logging, and alerts to support repeatable growth protocols.
- Developed Python and MATLAB analysis routines for STM/STS data, including constant-energy contour extraction and joint-density-of-states calculations for quasiparticle-interference studies.

Xi'an Jiaotong University

Mar 2022–Jul 2023

Research assistant, Yongchang Zhang group

- Simulated Rydberg-atom dynamics and photon-mediated interactions with MATLAB and Python to study quantum nonlinear optics.
- Built a discrete-time lattice simulation to examine two-qubit phase-flip dynamics and explore many-body regimes.

University of California, Berkeley

Oct–Dec 2021

Research assistant, Michael F. Crommie group

- Assisted with synthesis and STM characterization of graphene nanoribbons containing five-membered rings, supporting studies of their electronic states.

Publications

Zhen Zhu, Yudi Huang, Julian May-Mann, **Kaiming Liu**, Zheyu Wu, Shanta R. Saha, Johnpierre Paglione, Alexander G. Eaton, Andrej Cabala, Michal Vališka, Eduardo Fradkin, Vidya Madhavan.

[Evidence of intertwined pair density and charge density wave orders in UTe₂](#). *PNAS* 123 (29), e2602117123 (2026). [arXiv version](#).

Teaching

University of Illinois Urbana-Champaign

Aug–Dec 2023

Teaching assistant, PHYS 102

Led discussion and laboratory sections, held office hours, and provided feedback on assignments in electromagnetism and modern physics.

Selected Honors

China National Scholarship

2021

Xi'an Jiaotong University

Mathematical Contest in Modeling, Meritorious Winner

2021

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Technical Skills

Programming and computing: Python, NumPy, SciPy, JAX, MATLAB; Git; Linux/bash; parallel cluster computation.

Scientific methods: Probabilistic inference, Monte Carlo (MCMC/VMC), numerical optimization, forward-model validation, runtime profiling.

Experiment: Molecular beam epitaxy, STM/STS, ultrahigh-vacuum systems, feedback control, scientific data analysis.